LEVEL 1 – Basic Theory Questions (Board-level)

1. Basics of Matrices

1. Define a matrix.
2. What is the order of a matrix?
3. Define a row matrix with an example.
4. Define a column matrix with an example.
5. What is a square matrix?
6. What is a rectangular matrix?
7. Define a zero (null) matrix. Give an example.
8. Define a diagonal matrix.
9. Define a scalar matrix.
10. Define an identity matrix.
11. Define a symmetric matrix.
12. Define a skew-symmetric matrix.
13. Define an upper triangular matrix.
14. Define a lower triangular matrix.
15. Define a singleton matrix.
16. Give the conditions for two matrices to be equal.
17. How do you determine the order of a matrix?
18. Give examples of matrices of orders 1×1, 2×2, and 3×3.
19. Explain the difference between square and rectangular matrices.
20. Explain the difference between diagonal and scalar matrices.

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2. Matrix Operations

21. Define addition of matrices.
22. Define subtraction of matrices.
23. Explain scalar multiplication of a matrix.
24. Give an example of multiplying two matrices.
25. State the commutative property of matrix addition.
26. State the associative property of matrix addition.
27. State the distributive property of scalar multiplication over addition.
28. Define the transpose of a matrix.
29. State the properties of transpose of a matrix.
30. If A is a square matrix, what is the transpose of A^T?
31. Show with an example that $(A + B)^T = A^T + B^T$.
32. Can matrix multiplication be commutative? Give an example or counterexample.
33. Explain how to multiply a matrix by a scalar.
34. Explain the difference between matrix addition and multiplication.
35. Give an example where $AB \neq BA$.

36. Show that $(kA)^T = kA^T$.
37. What is meant by the identity matrix in matrix multiplication?
38. What is the zero matrix in addition of matrices?
39. Explain the effect of transpose on a diagonal matrix.
40. Show that the transpose of a symmetric matrix is the matrix itself.

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3. Determinants

41. Define determinant of a matrix.
42. What is the determinant of a 2×2 matrix?
43. What is a minor of an element in a matrix?
44. Define cofactor of an element.
45. State any two properties of determinants.
46. Explain expansion by minors (Laplace expansion).
47. Find the determinant of a diagonal matrix.
48. Find the determinant of a triangular matrix.
49. What is the determinant of the identity matrix?
50. Give an application of determinants in finding area of a triangle.
51. Give an application of determinants in solving linear equations.
52. Show that $\det(A^T) = \det(A)$.
53. Show that $\det(kA) = k^n \cdot \det(A)$, where A is $n \times n$.
54. Show that $\det(AB) = \det(A) \cdot \det(B)$.
55. Show that $\det(A^{-1}) = 1/\det(A)$.
56. Explain the effect of interchanging two rows on determinant.
57. Explain the effect of multiplying a row by a scalar on determinant.
58. Explain the effect of adding a multiple of one row to another on determinant.
59. Define singular and non-singular matrices.
60. State how determinant helps in finding invertibility.

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4. Inverse of a Matrix

61. Define the inverse of a matrix.
62. State the condition for a matrix to be invertible.
63. Explain the adjoint method to find inverse of a matrix.
64. Explain the row-reduction (Gaussian elimination) method to find inverse.
65. If A is invertible, show that $AA^{-1} = I$.
66. Show that $(A^{-1})^{-1} = A$.
67. Show that $(AB)^{-1} = B^{-1}A^{-1}$.
68. Find the inverse of a 2×2 matrix using formula.
69. Explain what happens if $\det(A) = 0$.
70. Show that $(A^T)^{-1} = (A^{-1})^T$.

5. Rank of a Matrix

71. Define the rank of a matrix.
72. Explain how to determine rank using echelon form.
73. Explain how to determine rank using determinants.
74. Give examples of matrices of rank 1, 2, and 3.
75. State the maximum rank of an $m \times n$ matrix.
76. How is rank used to solve linear systems?
77. Explain rank-nullity theorem (basic statement).
78. Determine the rank of a 2×2 zero matrix.
79. Determine the rank of a diagonal matrix with one zero element.
80. Explain consistency of a system using rank of matrices.

6. System of Linear Equations and Matrices

81. Represent a system of linear equations in matrix form.
82. Solve a 2×2 system using inverse method.
83. Solve a 2×2 system using Cramer's rule.
84. Solve a 3×3 system using Cramer's rule.
85. Solve a system using Gaussian elimination.
86. Explain conditions for a system to be consistent.
87. Explain conditions for a system to have a unique solution.
88. Explain conditions for a system to have infinitely many solutions.
89. Explain inconsistent system with an example.
90. Write augmented matrix of a system of linear equations.

7. Special Matrices and Their Properties

91. Define orthogonal matrix.
92. Show that for an orthogonal matrix A, $A^T A = I$.
93. Define idempotent matrix.
94. Give an example of idempotent matrix.
95. Define nilpotent matrix.
96. Give an example of nilpotent matrix.
97. Define trace of a matrix.
98. State two properties of trace.
99. Define elementary matrix.
100. Give an example of an elementary row operation using an elementary matrix.

8. Eigenvalues and Eigenvectors

101. Define eigenvalue and eigenvector.
102. Write the characteristic equation of a 2×2 matrix.
103. Explain how to find eigenvalues of a matrix.
104. Explain how to find eigenvectors of a matrix.
105. State properties of eigenvalues of a triangular matrix.
106. Explain diagonalization of a matrix.
107. Give an example of diagonalizable matrix.
108. Show that sum of eigenvalues = trace of matrix.
109. Show that product of eigenvalues = determinant of matrix.
110. Give an application of eigenvalues in differential equations.

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9. Applications of Matrices

111. Solve a system of linear equations using matrices.
112. Explain rotation transformation using 2×2 matrix.
113. Explain scaling transformation using matrix.
114. Explain reflection about x-axis using matrix.
115. Explain reflection about y-axis using matrix.
116. Solve a Markov chain problem using transition matrix.
117. Solve 2×2 Markov chain example.
118. Explain how matrices are used in computer graphics.
119. Give an example of matrix transformation in geometry.
120. Explain any real-life application of matrices in economics or science.